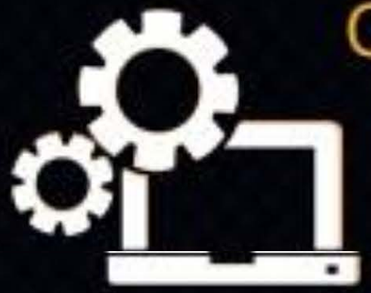
CS & IT ENGINEERING

DIGITAL LOGIC

Chapter:- Logic Gates



Lecture No. 1



By- CHANDAN SIR

Topics to be Covered



- Syllabus
- Weightage
- Reference Books
- NOT & AND GATE

CS & IT ENGINEERING

Digital Logic

Boolean algebra. Combination and sequential circuits. Minimization. Number representations and computer arithmetic (fixed and floating point).

CS & IT ENGINEERING

DIGITAL LOGIC	
Number of Questions	2 to 4
Marks	4 to 7
Frequently Asked Topics	Boolean algebra. Combinational and sequential circuits. Minimization. Number representation.

Reference Books



CJ BABA → NOTES



Book Name: Digital Design
Author: M. Morris Mano & Michael D. Ciletti
Publisher: Pearson Publisher

Book Name: Digital fundamental
Author: Thomas L. Floyd
Publisher: Pearson Publisher

Guidelines to Attend Live Class

- Attend the class with positive attitude.
- Punctuality is necessary.
- Follow the day-wise study plan.
- Attempt DPP daily after the schedule.
- Hold chat while attending the class. We will allow you to ask and put your question in the comment box.

Flip-Flop

JK Flip-Flop

IC → Integrated circuit

Jack Keilby → Papa of IC

1 element → 1958

2 element → 1960

4 element → 1962

8 element → 1964

Nobel Prize

Gorden E. Moore

1962

दुकान

INTEL

1965

Statement → Moore's Law

t.me/CJSIR

S.N.	Chapter	Topic
1	Logic Gate	NOT, AND, OR, NAND, NOR, X-OR, X-NOR
2	Minimization	Boolean algebra, K-MAP
3	Combination Circuit	Comparator, MUX, DE-MUX. Encoder, Decoder, HA, FA. H.S, F.S, Serial adder, Parallel adder. LACA. Multiplier, Code converter
4	Sequential Circuit	Latches, Flip-Flop, Register, COUNTERS
5	Number System	→ Base conversion → Magnitude Representation

Binary $\longrightarrow \{0, 1\}$

bit

Byte = 8 bit

Nible = 4 bit

$1 KH_z = 10^3 H_z$

$1 K = 10^3$

$1 K = 2^{10}$

$1 K bit = 2^{10} bit$

$= 1024 bit$

$1 Mb = 2^{20} bit$

$1 Gb = 2^{30} bit$

2 bit Binary

Binary	Decimal
0 0	0
0 1	1
1 0	2
1 1	3

	2 1
0	0 0
1	0 1
2	1 0
3	1 1

3 bits

	4 2 1
0	0 0 0
1	0 0 1
2	0 1 0
3	0 1 1
4	1 0 0
5	1 0 1
6	1 1 0
7	1 1 1

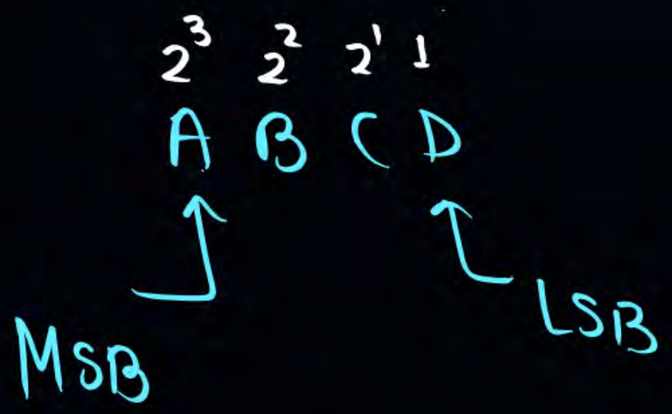
4 bit

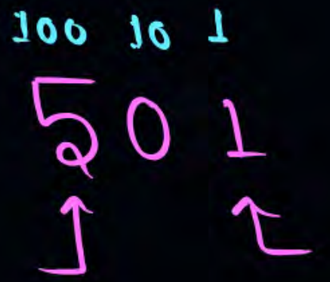
Decimal	Binary (8 4 2 1)
0	0 0 0 0
1	0 0 0 1
2	0 0 1 0
3	0 0 1 1
4	0 1 0 0
5	0 1 0 1
6	0 1 1 0
7	0 1 1 1

Decimal	Binary (8 4 2 1)
8	1 0 0 0
9	1 0 0 1
10	1 0 1 0
11	1 0 1 1
12	1 1 0 0
13	1 1 0 1
14	1 1 1 0
15	1 1 1 1

LSB → Least significant bit

MSB → Most significant bit



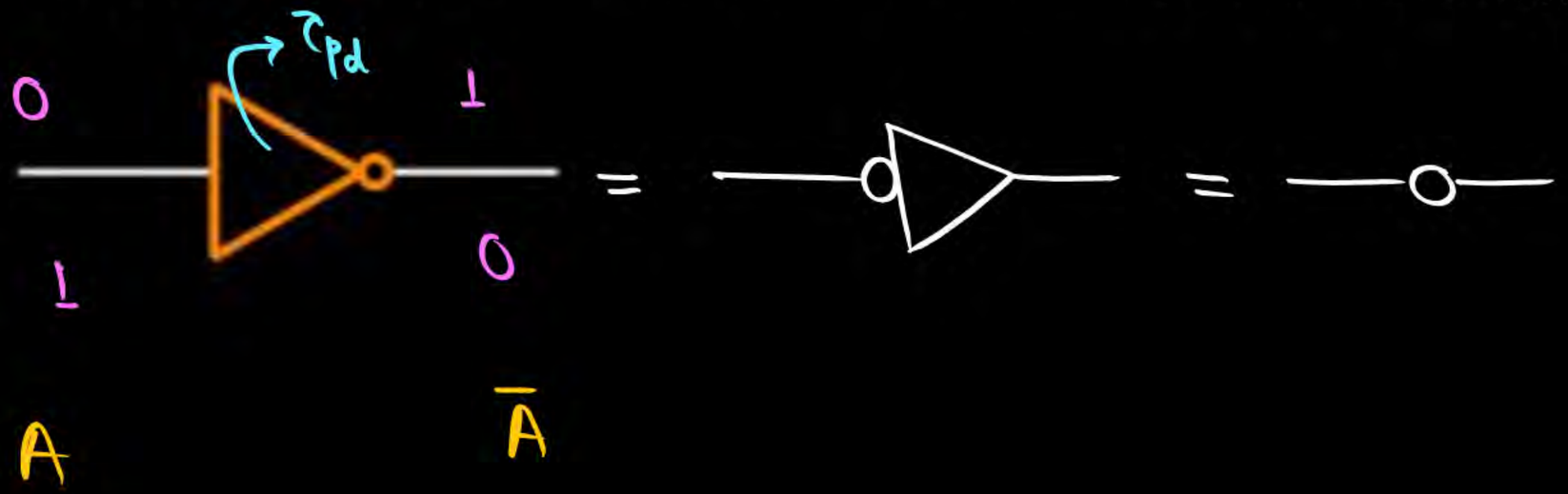


INVERTER GATE

(τ_{pd})

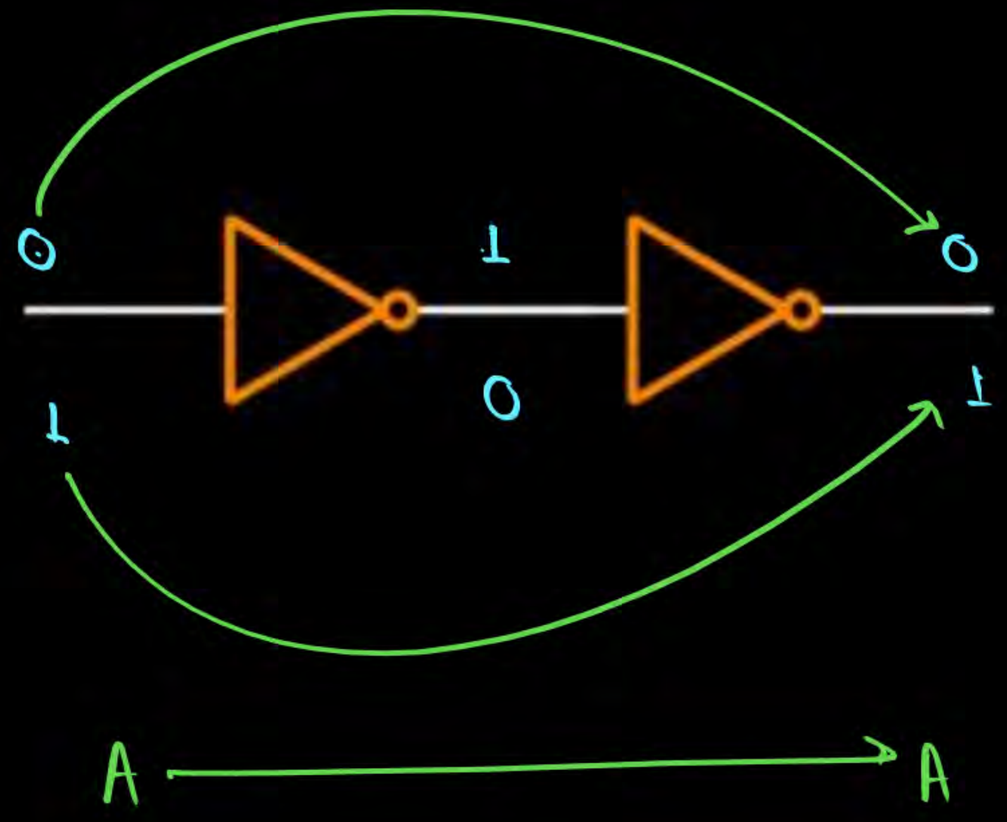
Propagation Delay :→ Time Required by the circuit to provide o/p after applying inputs.

1. NOT GATE [INVERTER, NEGATION, COMPLEMENT LOGIC]

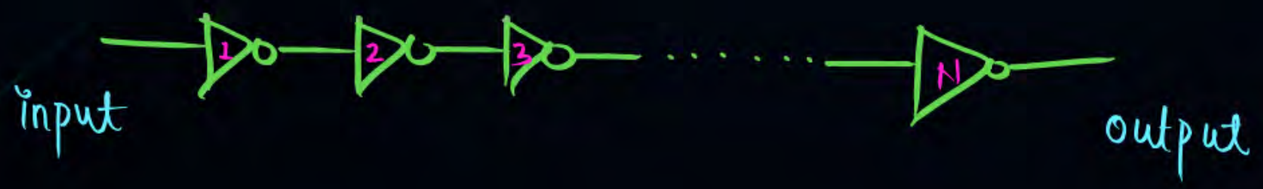


INVERTER GATE

1. NOT GATE



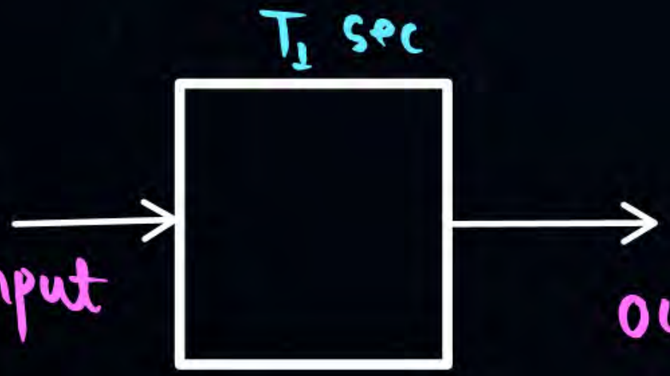
BUFFER



If N → odd ⟹ INVERTER

N → EVEN ⟹ BUFFER

Q



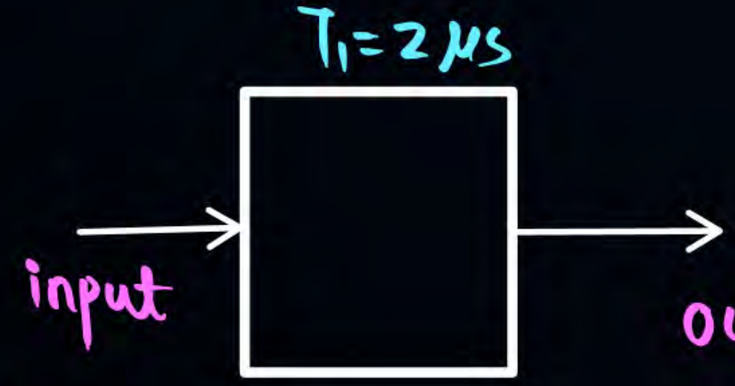
Delay = o/p Time − input Time

⇒ $t = 0$ sec — $t = T_1$ sec — Delay $= T_1 - 0 = T_1$

⇒ $t = 1$ sec — $t = T_1 + 1$ sec — Delay $= T_1 + 1 - 1 = T_1$

⇒ $t = T_i$ — $t = T_i + T_1$ — Delay $= T_i + T_1 - T_i = T_1$

Q



Delay = o/p Time − input Time

⇒ t = 0 sec — t = 2 μs — Delay = 2 μs − 0 = 2 μs

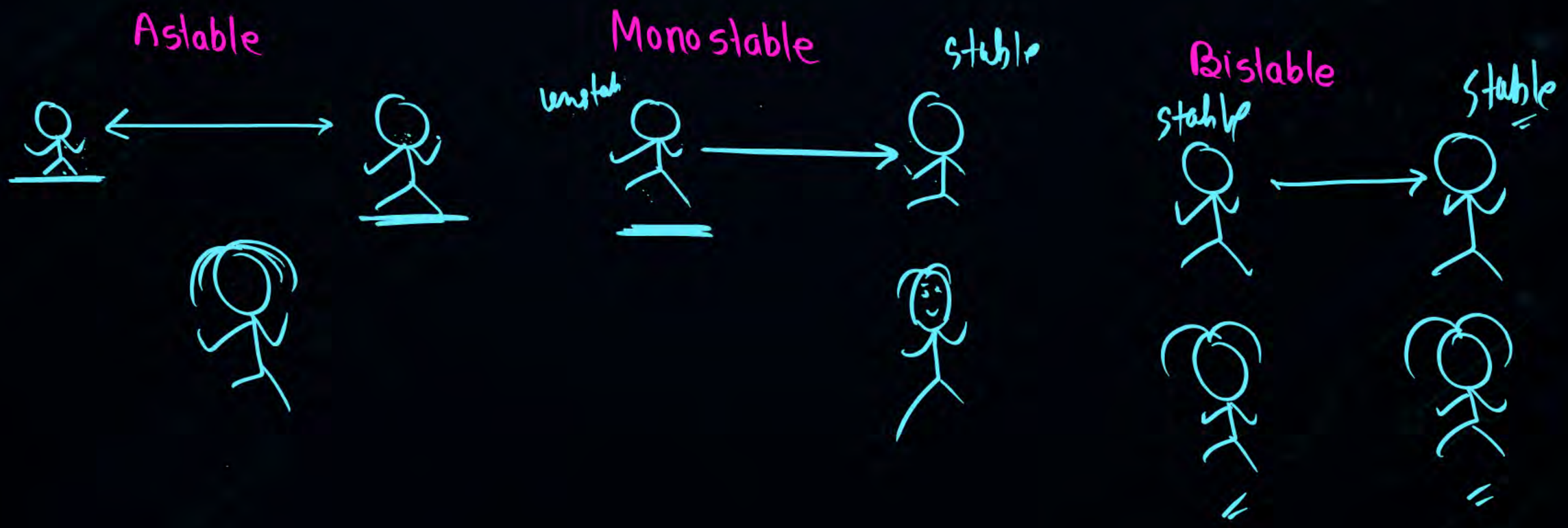
⇒ t = 1 sec — t = 3 μs — Delay = 3 μs − 1 μs = 2 μs

⇒ $t = T_i$ — $t = T_i + 2\mu s$ — Delay = $T_i + 2\mu - T_i = 2\mu$



$t=0$

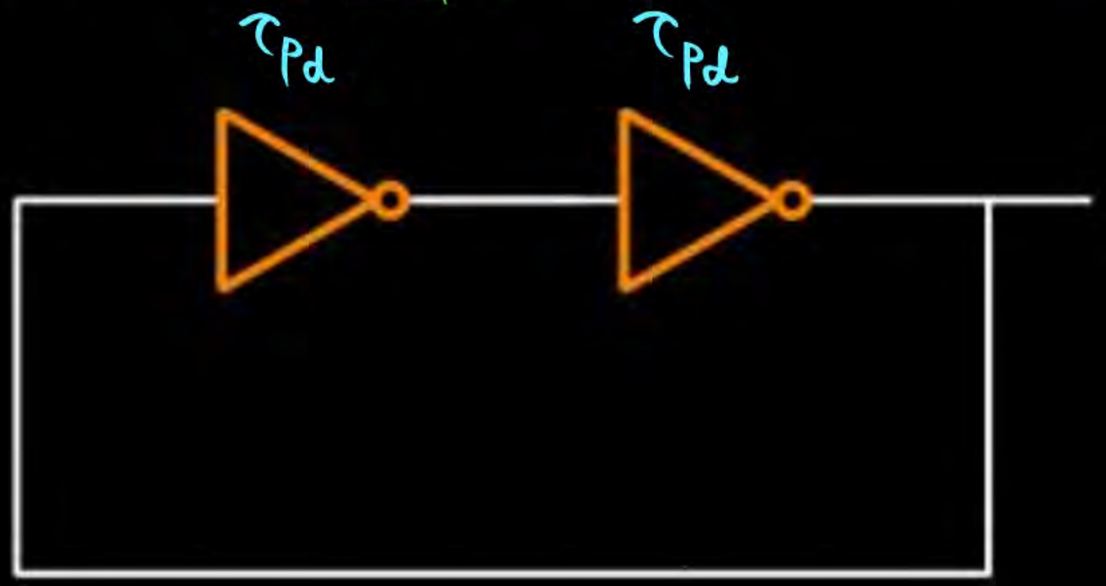
$t = 4\mu s$


Astable
Monostable
Stable
Bistable
Stable
Stable

INVERTER GATE

1. NOT GATE

When Even number of NOT GATE in Loop :-



HIGH

1 → 1 → 1 → 1 → ...

$2\tau_{pd}$ $2\tau_{pd}$ $2\tau_{pd}$ $2\tau_{pd}$

LOW

0 → 0 → 0 → 0 → ...

$2\tau_{pd}$ $2\tau_{pd}$ $2\tau_{pd}$ $2\tau_{pd}$

1) Basic Memory element
2) Bistable Multivibrator
3) AC generator ($f = 0$ Hz)

INVERTER GATE

1. NOT GATE

HW



?



2 Mint Summary

- NOT & AND GATE

Thank you DOSTO